

Tutorial sheet-1 - Solutions

① i) $\lim_{n \rightarrow \infty} \frac{n^2}{e^n} = \lim_{n \rightarrow \infty} \frac{2}{e^n} = 0 \Rightarrow \text{Converges}$

ii) $\lim_{n \rightarrow \infty} \frac{1-2n}{1+2n} = -1 \Rightarrow \text{Converges}$

iii) $\lim_{n \rightarrow \infty} \frac{n+3}{(n+3)(n+2)} = \lim_{n \rightarrow \infty} \frac{1}{n+2} = 0 \Rightarrow \text{Converges}$

iv) $\lim_{n \rightarrow \infty} 1 + (-1)^n = \begin{cases} 0, & \text{if } n \text{ is odd} \\ 2, & \text{if } n \text{ is even} \end{cases}$

$\therefore$ the limit of sequence does not exist and hence it is not convergent.

v) $\lim_{n \rightarrow \infty} \frac{(n-1)^2}{n-1} = \lim_{n \rightarrow \infty} n-1 = \infty \Rightarrow \text{diverges}$

vi) $\lim_{n \rightarrow \infty} \frac{n}{2^n} = \lim_{n \rightarrow \infty} \frac{1}{2^n \log 2} = 0 \Rightarrow \text{Converges}$

vii) $\lim_{n \rightarrow \infty} \frac{\sin n}{2^n} = 0 \Rightarrow \text{Converges}$

Since $0 \leq \sin n \leq 1 \Rightarrow 0 \leq \frac{\sin n}{2^n} \leq \frac{1}{2^n}$

$\downarrow$
Limit
value = 0

$\downarrow$
Limit
value = 0

By Sandwich theorem $\lim_{n \rightarrow \infty} \frac{\sin n}{2^n} = 0$

$$\text{viii)} \lim_{n \rightarrow \infty} \frac{\log n}{n^{\frac{1}{n}}} = \frac{\infty}{1} = \infty \Rightarrow \text{Diverges}$$

$$\text{ix)} \lim_{n \rightarrow \infty} \frac{n!}{n^n} \leq \lim_{n \rightarrow \infty} \frac{1}{n} = 0 \text{ and } \frac{n!}{n^n} \geq 0$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0 \Rightarrow \text{Converges}$$

$$\text{x)} \lim_{n \rightarrow \infty} \frac{5^n}{7^n} = \lim_{n \rightarrow \infty} \frac{5^n \log 5}{7} = \infty \Rightarrow \text{Diverges}$$

$$\text{xi)} \lim_{n \rightarrow \infty} \frac{\log n}{n} = \lim_{n \rightarrow \infty} \frac{1/n}{1} = 0 \Rightarrow \text{Converges}$$

$$\text{xii)} \lim_{n \rightarrow \infty} \left(\frac{n+1}{n-1} \right)^n = \lim_{n \rightarrow \infty} \frac{\left(1 + \frac{1}{n} \right)^n}{\left(1 - \frac{1}{n} \right)^n} = \frac{e}{e^{-1}} = e^2 \Rightarrow$$

Converges

$$\textcircled{2} \text{ soln } a_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!}$$

$$a_{n+1} = 1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!} + \frac{1}{(n+1)!}$$

$$a_{n+1} - a_n = \frac{1}{(n+1)!} > 0$$

$$\Rightarrow a_{n+1} > a_n \forall n$$

$\therefore \langle a_n \rangle$ is monotonically increasing sequence

$$\text{Also, } a_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!}$$

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$$= 1 + \frac{1}{1} + \frac{1}{1 \cdot 2} + \frac{1}{1 \cdot 2 \cdot 3} + \dots + \frac{1}{1 \cdot 2 \cdot 3 \cdot \dots \cdot n}$$

$$< 1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{n-1}} = 1 + \frac{1 - \frac{1}{2^n}}{1 - \frac{1}{2}}$$

~~no~~

$$= 1 + 2 \left(1 - \frac{1}{2^n}\right)$$

$$< 3$$

$$\therefore a_n < 3 \quad \forall n$$

$\therefore \langle a_n \rangle$ is bounded above sequence

Hence $\langle a_n \rangle$ is convergent.

③ Sol: Since n is a +ve integer, we have Binomial theorem,

$$a_n = \left(1 + \frac{1}{n}\right)^n = 1 + n \frac{1}{n} + \frac{n(n-1)}{2!} \frac{1}{n^2} + \dots + \frac{n(n-1) \dots 2 \cdot 1}{n!} \frac{1}{n^n}$$

$$= 1 + 1 + \frac{1}{2!} \left(1 - \frac{1}{n}\right) + \frac{1}{3!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) + \dots + \frac{1}{n!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{n-1}{n}\right)$$

changing n to $n+1$

$$a_{n+1} = 1 + 1 + \frac{1}{2!} \left(1 - \frac{1}{n+1}\right) + \frac{1}{3!} \left(1 - \frac{1}{n+1}\right) \left(1 - \frac{2}{n+1}\right) + \dots + \frac{1}{(n+1)!} \left(1 - \frac{1}{n+1}\right) \left(1 - \frac{2}{n+1}\right) \dots \left(1 - \frac{n}{n+1}\right)$$

Since $n+1 > n$

$$\Rightarrow \frac{1}{n+1} < \frac{1}{n}$$

$$\Rightarrow -\frac{1}{n+1} > -\frac{1}{n}$$

$$\Rightarrow 1 - \frac{1}{n+1} > 1 - \frac{1}{n}$$

Similarly, $1 - \frac{k}{n+1} > 1 - \frac{k}{n}$ for $k=2, 3, \dots, n-1$

$$\therefore a_{n+1} > a_n \quad \forall n$$

$\Rightarrow \langle a_n \rangle$ is monotonically increasing

$$\begin{aligned} \text{Also, } a_n &= 1 + 1 + \frac{1}{2!} \left(1 - \frac{1}{n}\right) + \frac{1}{3!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) + \dots \\ &\quad + \frac{1}{n!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{n-1}{n}\right) \end{aligned}$$

$$< 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{n!}$$

$$< 1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{n-1}}$$

$$= 1 + \frac{1(1 - \frac{1}{2^n})}{1 - \frac{1}{2}} = 3 - \frac{1}{2^{n-1}} < 3$$

$\Rightarrow \langle a_n \rangle$ is bounded above

$\therefore \langle a_n \rangle$ is convergent sequence.

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(4) Sol

$$a_1 = \sqrt{2}, \quad a_2 = \sqrt{2a_1} = \sqrt{2\sqrt{2}}$$

$$\sqrt{2} > 1$$

$$\Rightarrow 2\sqrt{2} > 2$$

$$\Rightarrow \sqrt{2\sqrt{2}} > \sqrt{2}$$

$$\Rightarrow a_2 > a_1$$

Suppose $a_n > a_{n-1}$ Then $2a_n > 2a_{n-1}$

$$\Rightarrow \sqrt{2a_n} > \sqrt{2a_{n-1}}$$

$$\Rightarrow a_{n+1} > a_n$$

$$\therefore a_{n+1} > a_n \quad \forall n$$

By Mathematical Induction, the sequence $\langle a_n \rangle$ is monotonically increasing.

Now $a_1 = \sqrt{2} < 2$

Suppose $a_n < 2$, then $2a_n < 4$

$$\Rightarrow \sqrt{2a_n} < 2$$

$$\Rightarrow a_{n+1} < 2$$

$\therefore$ By mathematical induction, $a_n < 2 \quad \forall n$

$\therefore \langle a_n \rangle$ is bounded above

Since $\langle a_n \rangle$ is monotonically increasing and bounded above, it is convergent.

Let $\lim_{n \rightarrow \infty} a_n = l$ then $\lim_{n \rightarrow \infty} a_{n+1} = l$

$$\therefore a_{n+1} = \sqrt{2a_n}$$

$$\Rightarrow a_{n+1}^2 = 2a_n$$

$$\text{as } n \rightarrow \infty, l^2 = 2l \Rightarrow l(l-2)$$

$$\Rightarrow l = 0 \text{ (or) } l = 2$$

But $l \neq 0$, since $\{a_n\}$ is monotonically increasing.

$$\sqrt{2} = a_1 < a_2 < a_3 < \dots$$

$$\Rightarrow \{a_n\} \text{ bounded below by } \sqrt{2} \Rightarrow \lim_{n \rightarrow \infty} a_n \geq \sqrt{2}$$

$$\text{Hence } l = 2$$

⑤ Sol $a_1 = 1, a_n = \sqrt{2 + a_{n-1}}$

$$a_2 = \sqrt{2 + a_1} = \sqrt{2 + 1} = \sqrt{3} > 1 = a_1$$

$$a_2 > a_1$$

Suppose $a_n > a_{n-1}$

$$\Rightarrow \sqrt{2 + a_n} > \sqrt{2 + a_{n-1}}$$

$$\Rightarrow a_{n+1} > a_n$$

By Mathematical induction, $a_{n+1} > a_n \forall n$

$\langle a_n \rangle$ is monotonically increasing

Also

$$a_1 = 1 < 2$$

Suppose $a_n < 2$, then $2 + a_n < 4$

$$\Rightarrow \sqrt{2 + a_n} < 2$$

$$\Rightarrow a_{n+1} < 2$$

By mathematical induction, $a_n < 2 \forall n$

$\therefore \langle a_n \rangle$ is bounded above

Since $\langle a_n \rangle$ is monotonically increasing and bounded above, it is convergent.

$$\text{Let } \lim_{n \rightarrow \infty} a_n = l \text{ then } \lim_{n \rightarrow \infty} a_{n+1} = l$$

$$\text{Now } a_{n+1} = \sqrt{2 + a_n}$$

$$\Rightarrow a_{n+1}^2 = 2 + a_n$$

$$\text{as } n \rightarrow \infty \quad l^2 = 2 + l \Rightarrow l^2 - l - 2 = 0$$

$$\Rightarrow (l-2)(l+1) = 0$$

$$\Rightarrow l = 2 \text{ (or) } l = -1$$

Since $a_n > 0 \forall n$, $l = \lim_{n \rightarrow \infty} a_n$ can't be -ve

$$\therefore l = 2$$

thus the sequence converges to the positive root of the equation $x^2 - x - 2 = 0$.

⑧ Try yourself.

⑨ soln let $a_1 > b_1$

for any two +ve numbers, the arithmetic mean is greater than the geometric mean

$$\therefore a_n > b_n \quad \forall n$$

Also, ~~$a_{n+1} = \frac{1}{2}(a_n + b_n)$~~ $a_{n+1} = \frac{1}{2}(a_n + b_n) < \frac{1}{2}(a_n + a_n) = a_n$

$$\Rightarrow a_{n+1} < a_n$$

$\therefore \langle a_n \rangle$ is monotonic decreasing

$$a_1 > a_2 > a_3 > \dots$$

again $b_{n+1} = \sqrt{a_n b_n} > \sqrt{b_n b_n} = b_n$

$$b_{n+1} > b_n$$

$\therefore \langle b_n \rangle$ is monotonic increasing

i.e, $b_1 < b_2 < b_3 < \dots$

Now $a_n = \frac{1}{2}(a_{n-1} + b_{n-1}) > \frac{1}{2}(b_{n-1} + b_{n-1}) = b_{n-1}$

$$a_n > b_{n-1} > b_{n-2} > b_{n-3} > \dots > b_1$$

$$\therefore a_n > b_1 \quad \forall n$$

⑤ This implies that the sequence $\{a_n\}$ is bounded below and being monotonic decreasing is convergent.

Again, $b_n = \sqrt{a_{n+1} b_{n+1}} < \sqrt{a_{n+1} a_{n+1}} = a_{n+1}$

$$b_n < a_{n+1} < a_{n+2} < \dots < a_1$$

$$\Rightarrow b_n < a_1 \quad \forall n$$

$\therefore \{b_n\}$ is bounded above and being monotonic increasing is convergent.

Now let $\lim_{n \rightarrow \infty} a_n = l$ and $\lim_{n \rightarrow \infty} b_n = l'$

Since $a_n = \frac{1}{2}(a_{n+1} + b_{n+1})$

$$\Rightarrow 2a_n = a_{n+1} + b_{n+1} \Rightarrow 2a_{n+1} = a_n + b_n$$

$$\text{as } n \rightarrow \infty \Rightarrow 2l = l + l' \Rightarrow l = l'$$

$\therefore \{a_n\}$ and $\{b_n\}$ converge to the same limit.

⑩ Try yourself

⑥ Sol:- Let $a_n = \frac{1}{(n+1)^2} + \frac{1}{(n+2)^2} + \dots + \frac{1}{(n+n)^2}$

$$> \frac{1}{(2n)^2} + \frac{1}{(2n)^2} + \dots + \frac{1}{(2n)^2} = \frac{n}{(2n)^2} = \frac{1}{4n}$$

$$a_n > \frac{1}{4n}$$

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Also, $a_n = \frac{1}{(n+1)^2} + \frac{1}{(n+2)^2} + \dots + \frac{1}{(n+n)^2}$

$$< \frac{1}{n^2} + \frac{1}{n^2} + \dots + \frac{1}{n^2} = \frac{n}{n^2} = \frac{1}{n}$$

thus $\frac{1}{4n} < a_n < \frac{1}{n}$

Since $\lim_{n \rightarrow \infty} \frac{1}{4n} = 0 = \lim_{n \rightarrow \infty} \frac{1}{n}$

By Sandwich theorem, we have $\lim_{n \rightarrow \infty} a_n = 0$

Hence the sequence $\langle a_n \rangle$ converges to 0.

⑦ soln- $a_n = \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{n+n}$

$$a_{n+1} = \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n} + \frac{1}{2n+1} + \frac{1}{2n+2}$$

$$a_{n+1} - a_n = \frac{1}{2n+1} + \frac{1}{2n+2} - \frac{1}{n+1}$$

$$> \frac{1}{2n+2} + \frac{1}{2n+2} - \frac{1}{n+1} = \frac{2}{2n+2} - \frac{1}{n+1} = 0$$

$$a_{n+1} - a_n > 0 \quad \forall n$$

$$\Rightarrow a_{n+1} > a_n \quad \forall n$$

$\Rightarrow \langle a_n \rangle$ is monotonically increasing

Also, $a_n = \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{n+n}$

$$< \frac{1}{n+1} + \frac{1}{n+1} + \dots + \frac{1}{n+1} = \frac{n}{n+1} < 1 \quad \forall n$$

$$\Rightarrow a_n < 1 \quad \forall n$$

$\Rightarrow \{a_n\}$ is bounded above

Since $\{a_n\}$ is monotonically increasing and bounded above, it is convergent.

Now
$$a_n = \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n}$$

$$> \frac{1}{2n} + \frac{1}{2n} + \dots + \frac{1}{2n} = \frac{n}{2n} = \frac{1}{2} \quad \forall n$$

$$a_n > \frac{1}{2} \quad \forall n$$

$$\therefore \frac{1}{2} < a_n < 1 \quad \forall n$$

Hence, $\frac{1}{2} \leq \lim_{n \rightarrow \infty} a_n \leq 1$.

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